Hard Negative Mining for Knowledge Graph Link Prediction: A Controlled Study on FB15k-237

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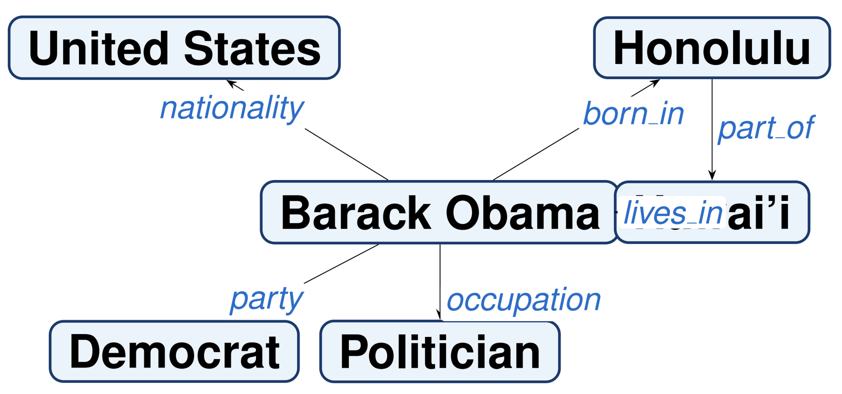
Can training on smarter mistakes teach an AI to reason better over real-world facts?

What is a Knowledge Graph?

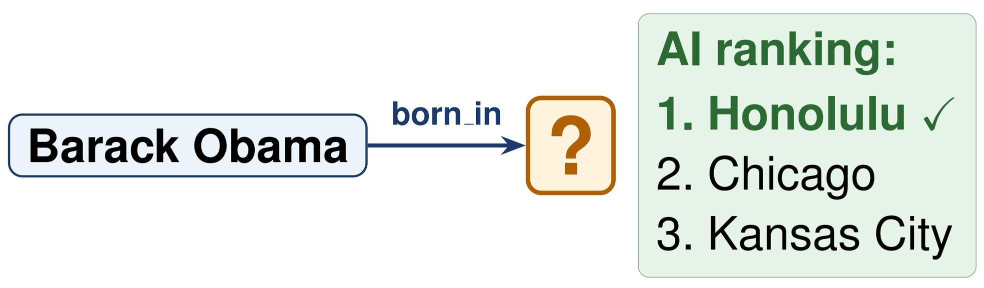
A **knowledge graph** (KG) is a structured network of real-world facts. Each fact is stored as a simple triple: **(subject, relation, object)**. Google, Siri, Netflix, and Wikipedia all rely on knowledge graphs behind the scenes.



We use FB15k-237: 272,115 training facts, 14,505 entities, 237 relation types, sourced from Freebase.

Predicting Missing Links

KGs are always *incomplete*: facts are missing. **Link prediction** is the task of automatically completing the graph by ranking all candidate answers for a given query.



We measure quality with **MRR** (Mean Reciprocal Rank): if the right answer ranks 1st the score is 1.0; if 2nd, it is 0.5; and so on. Higher MRR means better predictions.

Teaching AI with Wrong Answers

Models learn by contrasting correct facts with *fabricated wrong facts* (negatives). The quality of these negatives determines how much the model learns at each training step.

RANDOM NEGATIVES (standard)

Obama → born_in → *Ice cream*

Obama → born_in → *Mt. Everest*

Obama → born_in → *Atlantic Ocean*

Obviously wrong. The model stops improving after a few epochs.

HARD NEGATIVES (our method)

Obama → born_in → *Chicago*

Obama → born_in → *Kansas City*

Obama → born_in → *Honolulu*[*]

Plausible but wrong. The model keeps improving.

[*]Honolulu *is* the right answer here, which makes it the hardest possible negative to distinguish.

Our two-stage pipeline:

1. Generate 64 random candidate wrong answers (Bernoulli corruption).
2. Select the 8 most informative ones using a **strategy**: **Random** (control), **Hard** (top-scoring by the model), or **Mixed** (blend of both). Zero extra compute.

Key Result

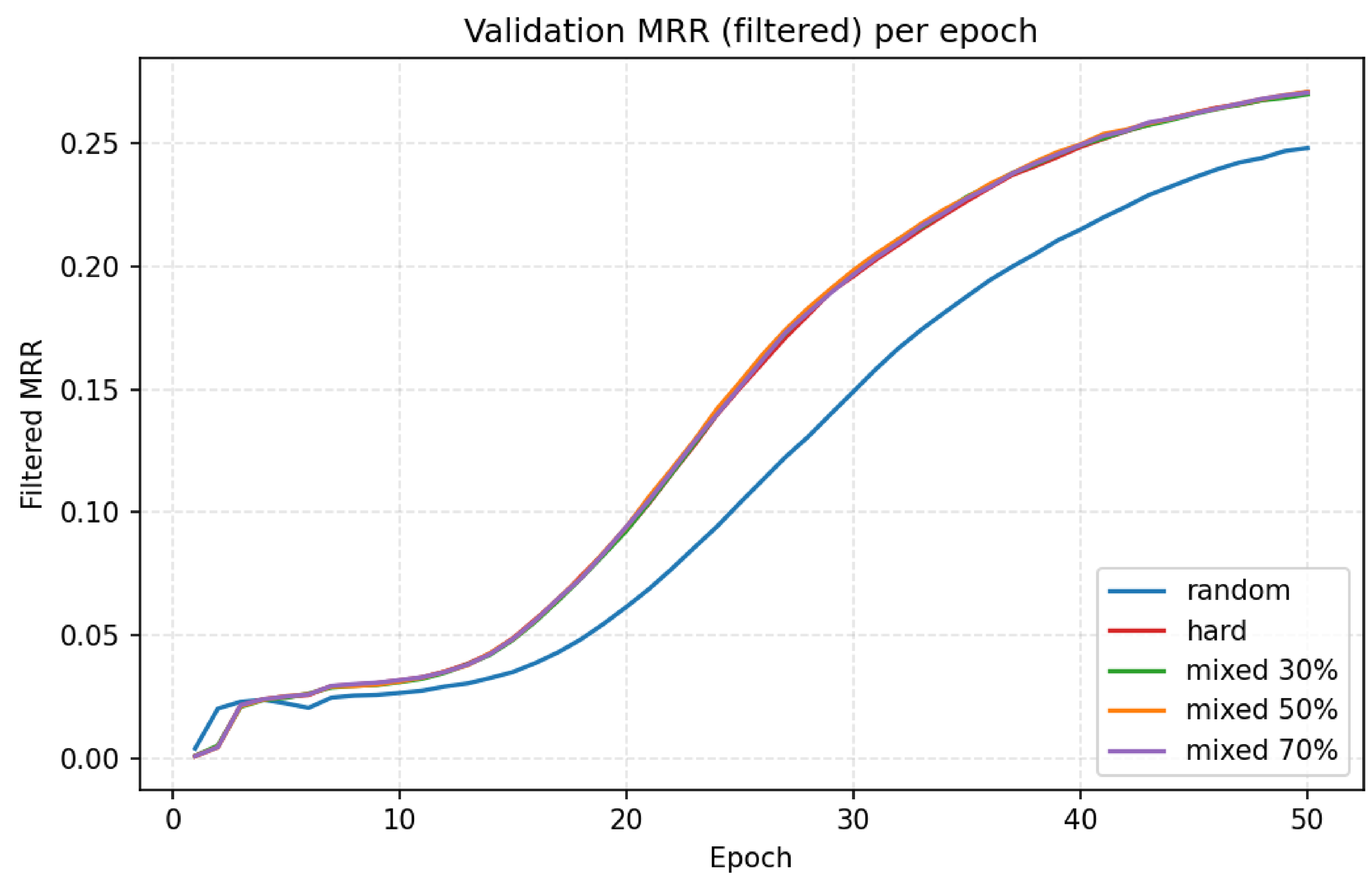
+10.4 %

accuracy improvement (MRR) using Hard negatives compared to standard random negative sampling

Strategy	MRR	H@1	H@3	H@10
Random (baseline)	0.270	0.189	0.300	0.428
Hard	0.298	0.217	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	0.328	0.460
Mixed 70%	**0.299**	**0.217**	**0.328**	**0.461**

Mixed 30% hard recovers 95% of the gain at zero extra cost.

How It Plays Out Over 50 Epochs



Hard and mixed strategies **pull ahead from random after epoch 15** and reach a consistently higher plateau, with no training instability.

Where Does the Gain Come From?

Hard negatives benefit *every* part of the dataset, but the effect is strongest for **rare relations**, where random negatives are least informative.

+3.6 pp
Rare relations
0.382 → 0.418

+3.0 pp
Medium relations
0.332 → 0.362

+2.8 pp
Frequent relations
0.257 → 0.285

Take-Aways

- **Smarter negatives, free upgrade:** the hard negative strategy requires no extra computation — only a smarter selection from the same pool.
- **You don't need 100% hard:** Mixed 30% (roughly 2 hard examples per 8) recovers 95% of the accuracy gain with greater training stability.
- **Rare facts benefit most,** exactly where knowledge graph completion is hardest and most valuable in practice.
- **Next steps:** dynamically adjust the hard fraction during training; scale experiments to Wikidata-5M.

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